

Hybrid Xception-LSTM Model for Explainable Mango Ripeness Classification Using Grad-CAM

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Abstract—The quality of fruits is one of the main concerns for consumers. There are international market standards that need to be considered when fruit goes to general consumers. The effectiveness and quick classification of the Mango's ripeness stage are the most decisive factors in determining its quality. Still, the assessment of the maturity level of the Mangoes is done manually. This is a very time-consuming and costly process, and it is one of the key challenges in the agriculture sector. In this research, a hybrid classification model is proposed for ripeness detection. The dataset contains 6000 images in total and includes six classes of ripeness labels. In this study, a hybrid classification model combining LSTM with the Xception model was used. Grad-CAM visualizations were utilized to highlight biologically relevant regions, such as color and texture variations, that guided the predictions. With this approach, the results obtained were outstanding with an accuracy of 98%. It ensures that the ripeness detection is working correctly.

Index Terms—Mango ripeness detection, LSTM, Xception, Deep learning.

I. INTRODUCTION

The Mango is considered to be among the most famous fruits globally, with almost 16 percent of the world's fruit production according to the Food and Agriculture Organization (FAO) [1].

Mangoes contain high levels of vitamins, minerals as well as antioxidants and are commonly eaten in their fresh state or processed in different forms such as juice, smoothies, dessert, and sauce. Nevertheless, the level of consumer satisfaction, shelf life, and market value is very much dependent on the ripeness of the fruit. As an example, overripe mangoes have more sugar in them, which cannot be used by diabetic patients [2].

Conventionally, ripeness determination is done through manual examination in terms of color, firmness, and aroma. Although this method works well on a small scale, the process is subjective, labor-intensive, and inconsistent, especially in commercial supply chains that process large quantities of fruit. This has necessitated the desire to have automated, cost-effective, and reliable systems of ripeness detection to enhance quality control, minimize post-harvest losses, and optimize the packaging and distribution processes [3].

The latest progress in the field of deep learning has provided encouraging outcomes in the agricultural field, specifically in the area of fruit classification. CNNs and transfer learning based on VGG, ResNet, DenseNet, and Xception architectures have been employed successfully in image classification tasks. However, lots of research works are based on rather small datasets, which restricts generalization. To overcome them, the present study presents a hybrid deep learning framework with Long Short-Term Memory (LSTM) networks and Xception architecture for detecting mango ripeness. A few existing datasets were combined to create a dataset of more than 6,000 images on six levels of ripeness. The data set can be used to train and evaluate more reliably than other past studies, which had small samples.

The key findings of this work are:

- 1) Growing a large and diverse mango ripeness dataset.
- 2) Adoption of a hybrid LSTM + Xception architecture to classify ripeness.
- 3) Comprehensive performance testing of the model to be potentially applicable in the agriculture and food industries.

The novelty of the given research is based not only on the

integration of Xception and LSTM models but also on the reinterpretation of the idea of temporal modeling on static fruit images. In the given model, the feature maps obtained in Xception are viewed as pseudo-sequences of spatial activations, which allows the LSTM to visualize inter-regional contextual dependencies that replicate the biological mechanism of mango ripening. This allows the network to find subtle spatial gradients in color and texture, which are not effectively modeled when a conventional CNN is used. Furthermore, this study presents a six-stage classification process, which includes Unripe, Ripe, Early Ripe, Partially Ripe, Ripe, Overripe, and Perished. For the study, a balanced and standardized dataset was created by integrating two publicly accessible sources, Kaggle and Mendeley, with consistent relabeling and augmentation. The addition of Grad-CAM visualization helps identify biologically relevant parts of the images that affect classification decisions, ensuring greater transparency and credibility in the case of agricultural implementation, making it more interpretable.

II. LITERATURE REVIEW

A study by Tripathi and Maktedar used a unique lion-assisted firefly approach with CNN for mango ripeness classification on a dataset with 748 images with 3 classes. The highest accuracy was 94.5% on classifying Ripe-Unripe mangoes [4].

Tiwari et al. (2023) used federated learning to classify mango ripeness on a 975-image dataset with 3 classes by using pretrained frameworks of deep learning, which simulate 3-7 clients. InceptionNet was the best performing with an accuracy of 98.75% [5].

Bhole and Kumar (2020) created a non-destructive system of grading 'Kesar' mangoes into 3 classes based on transfer learning using SqueezeNet on RGB and thermal images. The RGB and thermal accuracy were 93.33% and 92.27%, respectively [6].

Salazar-Campos et al. (2025) developed a lightweight single-layer CNN model for classifying 'Kent' mango maturity stages per Peruvian Technical using 603 labeled images. They achieved 96.04% accuracy [7].

Guru and Nandini (2024) introduced a Vision Transformer model to evaluate the ripeness of 'Alphonso' mangoes using 979 images with 4 classes. They used ViT Base 16, ViT Large 16, and ViT Huge 14 pre-trained weights. ViT_Huge_14 achieved 92.78% accuracy [8].

Sikder et al. (2025) compared deep learning, transfer learning, and traditional ML to predict mango ripeness using a custom dataset, and feature extraction using CNN and VGG16, followed by ML classification. Gradient Boosting CNNs recorded the best 96.28% accuracy, higher than VGG16 and normal ML [9].

The article by Wu et al. (2020) focused on the issue of automatic quality grading of mangoes with deep learning, based on a dataset of 6,400 images with three classes. They evaluated pretrained CNNs with Mask R-CNNs on background removal and convolutional autoencoder-classifiers. VGG16

using transfer learning had the highest test accuracy of 83.5% as compared to other models [10].

A hybrid CNN-SVM model was proposed by Rasool et al. 2025 to classify the ripeness of mangoes on the basis of 975 images. CNNs are used to extract hierarchical features of an image, and SVMs are used to classify them, with 98.53 percent accuracy [11].

III. METHODOLOGY

Figure 1 shows the system diagram of the pipeline for mango ripeness classification with the help of a hybrid deep learning approach. The first step involves obtaining input images and preprocessing, namely, resizing them to 224x224 pixels and augmenting them to enhance generalization. Spatial features are then extracted through the Xception network, which makes image representations rich. These properties are input into the LSTM layer, which learns sequential patterns and learns ripening dynamics. With the help of a Softmax layer, the classifications of features combined are divided into six categories of ripeness. Also, a Grad-CAM visualization package makes the model more interpretable by producing heatmaps that summarize areas of discrimination affecting classification.

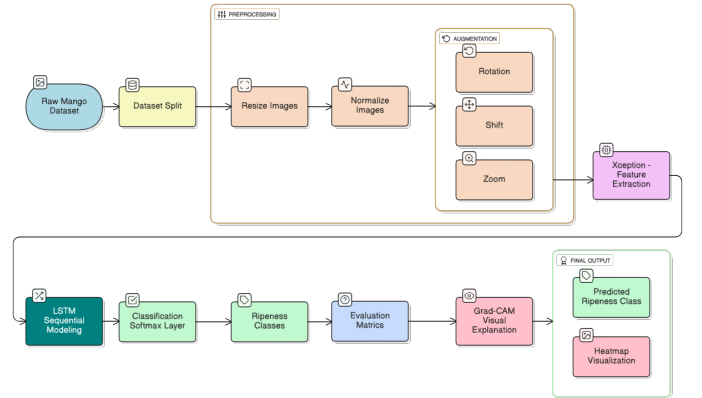


Fig. 1. System Overview

A. Dataset Preparation

The dataset was made by combining two different datasets, which were taken from Mandeley [12] and Kaggle [13]. The dataset from Mandeley contained 1080 images with two classes, which are overripe and perished. And the dataset from Kaggle had 951 images with four different classes, such as unripe, early ripe, partially ripe, and ripe. To balance the dataset, data augmentation such as rotation, shifting, and zooming was performed. After the augmentation was done, each class had 1000 images, thus creating a well-balanced dataset with 6000 images in total. Next, the dataset was split into 80-10-10 ratios for training, testing, and validation. Training was done with 4800 images, testing with 600 images, and validation with 600 images. Figure 2 shows a sample of the dataset being used in this study, which contains images of all six classes.

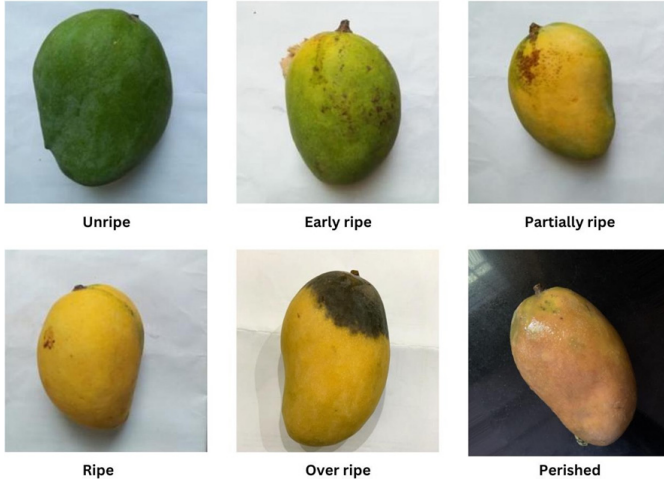


Fig. 2. Dataset Samples

The combination of two distinct sources adds variety in terms of lighting, background, camera angles, and settings of acquisition. All images were rescored rather than compromising the consistency of labels, on a well-defined six-stage ripeness scheme, so that there was uniform labelling across sources. This combination approach minimizes overfitting, increases the robustness of the model and exposes the network to a wider range of visual patterns and hence makes generalization in real world situations more reliable.

B. Data Preprocessing

Preprocessing data ensures its quality and applicability for further algorithmic analysis. The dataset was preprocessed using a standardized preprocessing pipeline in order to prepare it to be trained. Mango images were all downsampled to 224×224 pixels to fit in the input of the Xception network and scaled to the range $[0,1]$ by dividing pixel values by $1/255$. Controlled data augmentation was used to reduce overfitting and enhance generalization, with random 30° rotations, horizontal and vertical displacements of 10 percent, and zoom variations of 10 percent.

C. Fusion of Xception & LSTM

The proposed method, as shown in Figure 3 uses a hybrid Xception and LSTM model of mango ripeness classification. The architectures are based on the Xception network, which is pretrained on ImageNet and obtains the fine-grained visual features through depthwise separable convolutions. The obtained features are restructured to their sequential values, and the LSTM layer using the ReLU activation and the batch normalization is used to extract the contextual features across regions of the fruit surface. The resulting feature maps are flattened and fed through fully connected layers and a Softmax classifier, which dedicates each input image to its ripeness category. Adam optimizer is used with an initial learning rate of 1×10^{-4} , categorical cross-entropy loss, a batch size of 64, and a maximum of 50 epochs to train. To avoid

overfitting, early stopping and reduction of the learning rate are used. Although Xception is effective in space hierarchy, LSTM addition helps to improve time dependency modeling to enable the detection of minor changes related to ripening and resistance to inter-class similarity.

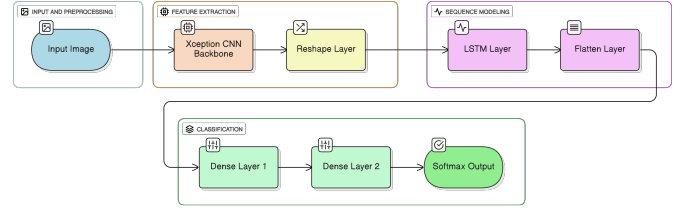


Fig. 3. Architecture of Xception + LSTM

D. Spatial-Context LSTM Architecture

Compared to the traditional use of LSTM networks in the temporal context (like video sequences or time-series prediction), the model proposed in the paper assumes the use of LSTM to learn spatial relations in a single static image. The rationale behind this design is that the process of mango ripening is non-uniform as it advances in various parts of the surface of the fruit. An example of this is the stem area, which tends to be green as compared to the belly area, which tends to turn yellow earlier. Thus, the process of ripening is progressive, with a color change, and is not uniform. The detection of the stage of mango ripeness is not just based on the fact that it has certain colors, but also on the location of the colors. The suggested LSTM module represents such region-specific correlations, where the network can explain localized variations that indicate various stages of ripeness.

Spatial-Context LSTM: Regional Dependency Modeling

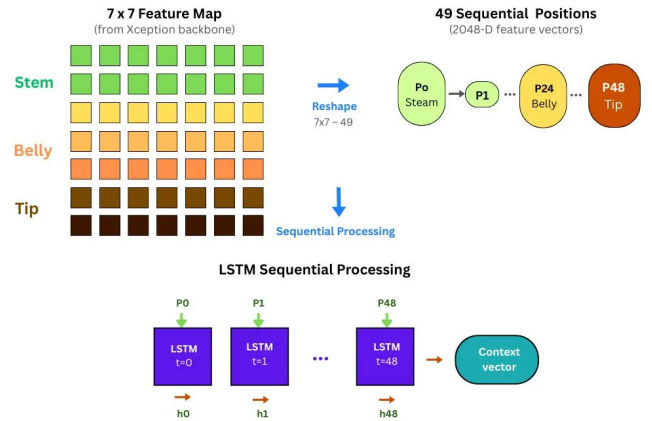


Fig. 4. Spatial-Context LSTM for modeling regional dependencies across the mango surface.

As shown in Figure 4 The feature output of the Xception backbone is a $7 \times 7 \times 2048$ feature tensor that is reshaped into a sequence of 49 spatial positions, each of which is modeled by a feature vector of 2048 dimensions. The sequence starting from the top to the bottom of the picture is in a row-wise order,

which shows the vertical orientation of mangoes and therefore allows the LSTM to understand the contextual interactions between neighboring and distant parts of the fruit surface. The LSTM has hidden states that represent local information available in the current area and the contextual information available in the previous areas, which in turn models the spatial evolution of ripening.

Let $F \in \mathbb{R}^{7 \times 7 \times 2048}$ denote the feature map extracted from the Xception network. This feature map is reshaped into a sequence

$$S = \text{Reshape}(F) \in \mathbb{R}^{49 \times 2048},$$

where S_i represents the feature vector corresponding to spatial position i ($i = 0, 1, \dots, 48$).

The LSTM processes this sequence as

$$h_t = \text{LSTM}(S_t, h_{t-1}),$$

where h_t encodes both the features of the current region and the contextual information from preceding spatial locations.

In such a way, the network can deduce a biologically significant spatial pattern, which may include upper parts of the image staying mostly green, whereas the middle parts of the image turn yellow due to the approaching ripeness, or the lower parts of the image may show brown discolorations due to the overripeness.

E. GradCam

Gradient-weighted Class Activation Mapping (Grad-CAM) was to be used to enhance the interpretability of the proposed deep learning framework. Grad-CAM is a visualization method that reveals the sections of an input image that have the greatest influence on the decision made by the model. With the help of the gradients of the target class into the final convolutional layer, Grad-CAM generates a class-discriminative heatmap, the spatial areas that contribute to the classification decision [14]. Grad-CAM was used on mango images in this study to produce a visual explanation of each of the ripeness classes predicted. This not only serves to support the fact that the model is targeting biologically relevant parts of the fruit but also adds to the openness and reliability of the classification system.

IV. RESULT AND ANALYSIS

The proposed Xception+LSTM model is evaluated using standard metrics. Accuracy is the total assessment of the ratio of the correctly classified samples. In contrast, precision and recall assess the specific class reliability by defining false positives and false negatives, respectively. The harmonic mean of both the precision and recall gives a balanced measure of the performance, the F1-score.

The classification report in Table I and Figure 5 shows the soundness of the proposed model in all six mango ripeness stages. Precision, recall, and F1-scores are all high, with most of the values being greater than 0.95. Interestingly, the stage0 (Unripe) reported the highest recall of 1.00, indicating that all the unripe samples were correctly identified with no false

TABLE I
PERFORMANCE METRICS FOR DIFFERENT STAGES OF UNRIPE CLASSIFICATION.

Class Name	Precision	Recall	F1-score
Stage_0 (Unripe)	0.91	1.00	0.95
Stage_1 (Early Ripe)	1.00	0.90	0.95
Stage_2 (Partially Ripe)	0.99	0.99	0.99
Stage_3 (Ripe)	0.99	1.00	1.00
Stage_4 (Overripe)	1.00	0.99	0.99
Stage_5 (Perished)	0.99	0.99	0.99

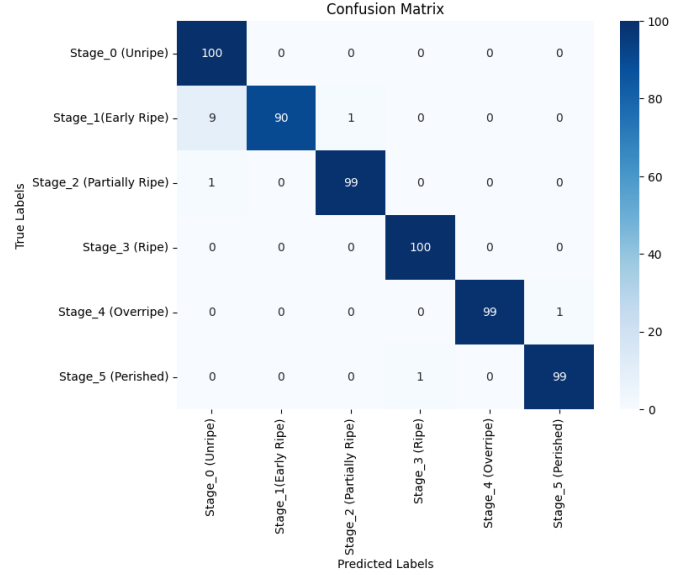


Fig. 5. Confusion matrix of the hybrid Xception-LSTM model

negatives. Likewise, Stage_3 scored high in recall and F1, which indicates that the model is able to detect discriminative features at intermediate ripeness levels. The model has a high discriminative accuracy at all six stages of ripeness with an almost perfect classification accuracy.

Stages 2, 4, and 5 had a moderate accuracy and recall (0.99-1.00), which means that the model does not misclassify one stage more than others. The very high accuracy on all measures confirms that Xception+LSTM is able to recognize the indistinguishable differences between neighboring ripeness levels, which are hard to detect by human eyes and simple deep learning models.



Fig. 6. (a) The train and validation loss of Xception-Lstm is against the number of epochs. (b) Train and Validate the accuracy of Xception-Lstm against the number of epochs

Training of the model was limited to a maximum number of 50 epochs, although the early stopping was triggered at epoch 23, where the validation performance had reached its stabilization, thus eliminating unnecessary computation and reducing overfitting. The training loss curve and the validation loss curve Figure 6(a) indicate that the loss curve decreases sharply at the first few epochs and stabilizes at low values after that. Equally, the training and validation accuracy curve Figure 6(b) illustrates a quick improvement that reaches an accuracy of more than 97 percent at about epoch 12, and then at the same time, the accuracy remains steady thereafter. The nearly parallel training and validation curves indicate that the proposed Xception-LSTM model generalizes well to unseen data. In order to further confirm the efficacy of the suggested

TABLE II
SCORE COMPARISON WITH PRETRAINED MODEL

Model	Precision	Recall	F1-Score	Accuracy
Xception+LSTM	98%	98%	98%	98%
Xception	94%	94%	94%	94%
CNN	94%	94%	94%	94%
ResNet50	92%	92%	92%	92%
VGG19	95%	95%	95%	95%

solution, the outcomes have been compared with four popular deep learning benchmarks, such as Xception, CNN, ResNet50, and VGG19. As shown in Table II, our research showed that the proposed hybrid model performed well, with an accuracy of 98%, outperforming all pretrained models. The standalone Xception model's performance is also less than the proposed hybrid model.

TABLE III
COMPARISON WITH EXISTING PAPER

Authors	Method	Dataset	Accuracy
Tripathi and Maktedar	Lion assisted firefly approach with CNN	748 images with 3 classes	94.5%
Tiwari et al.	InceptionNet	975 images with 3 classes	98.75%
Sikder et al.	CNN with Gradient Boosting	3900 images with 3 classes	96.28%
Rasool et al.	Hybrid CNN-SVM	975 images with 3 classes	98.53%
This study	Xception+LSTM	6000 images with 6 classes	98%

In order to evaluate the contribution of this work in the general research environment, the performance was compared to some of the latest studies in the field of mango ripeness detection, as shown in Table III: Based on this comparison, the proposed Xception+LSTM has a performance equal to or better than the best-reported accuracies. Although Tiwari et al. (98.75%) and Rasool et al. (98.53) have a little more accurate results, their datasets were smaller and very imbalanced, which makes them hard to generalize. By contrast, this paper utilizes a larger and more evenly distributed dataset (6,000 images) using six stages of ripeness, which provides a finer granularity compared to most prior works, which will often only use three stages of ripeness.

Figure 7 shows Grad-CAM images of mango ripeness recognition at six stages. The figure indicates that the suggested Xception-LSTM framework is effective in the localization of discriminative regions among the ripeness categories: Unripe (Stage 0), Early Ripe (Stage 1), Partially Ripe (Stage 2), Ripe (Stage 3), Overripe (Stage 4), and Perished (Stage 5). The actual and predicted outputs of each stage are indicated, as well as the Grad-CAM heatmap and its overlay. It can be seen in the visualizations that unripe and early ripe stages are characterized by activations at the central areas of the skin, whereas partially ripe and ripe stages are characterized by yellow pigmentation over a wider region. The results of over-ripe fruits give emphasis on dark spots and irregular pigmentation, but perished fruits emphasize the shriveled and discolored regions of the fruit, and this proves that the model is biologically relevant and allowable in the process of making reliable decisions in agriculture.

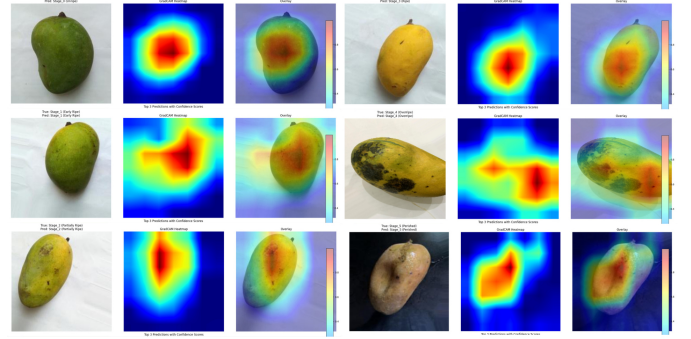


Fig. 7. GradCAM Images of different ripeness stages

V. CONCLUSION

The designed hybrid model Xception-LSTM was found to reach 98% accuracy, precision, recall and F1-score, which is greater than the conventional CNNs and corresponds to the highest reported values. In contrast to the previous literature, which could only consider a smaller number of classes or smaller datasets, this paper classifies six ripeness stages with a large-scale dataset and proves to be resistant to long ripening periods and inter-class performances. The combination of the Xception-based spatial feature extraction and LSTM-based temporal modeling is a good way to reflect the fine-tuning of ripening changes, and the Grad-CAM can be used to explain the image, indicating the parts of the body that are biologically significant and make a difference in the classification decisions.

The future studies will involve scaling the data with multisensory and multispectral samples in order to retrieve more detailed aspects and create lightweight variations of the models that can run on mobile and edge devices. To increase the potential of the framework in smart agriculture and automated supply chain management, it might be necessary to incorporate real-time monitoring and test the framework in a variety of environmental conditions.

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